SOFTWARE DESIGN DOCUMENTATION

PROJECT: Traduce Señas

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Introduction

The design document translates the requirements and analysis of Traduce Señas into concrete construction decisions. It brings together four views: the graphical interfaces the user sees, the architectural model that organizes the components, the design patterns applied in the code, and the data model that supports the system's information.

The decisions described here correspond to the code in the frontend and backend repositories (frontend 1.3.1 and backend 1.4.0) and to the high-fidelity prototypes in the documentation repository.

Graphical interface design

Design principles

Because the main users are deaf people, the design is based on accessibility (W3C, 2018) and Nielsen's usability principles (1994):

- **Nothing by sound alone:** every audio output has a visible text equivalent.
- **Sufficient contrast:** 4.5:1 for normal text and 3:1 for large text, in light and dark themes.
- **Large touch targets:** minimum 48 × 48 dp; primary actions are 56 dp high.
- **One primary action per view**, so that the user always knows what to do next.
- **Readable text:** body text of 18 px on mobile and 16 px on desktop; no data below 14 px.

Design system

Table 1
Design system elements

Element	Definition
Themes	Light (default) and dark; accent color purple (brand) or green, saved on the device
Typography	Nunito Sans. H1 title 32 px, H2 24 px, H3 20 px, body 18/16 px, buttons 18 px in bold
Spacing	Scale of 4, 8, 12, 16, 20, 24, 32, 40, 48, and 72 px; side margins of 16 px on mobile and 24 px on desktop
Borders	Radius of 12 px on buttons and fields, 18 px on cards
Breakpoints	Mobile < 768 px, tablet 768–1023 px, desktop ≥ 1024 px
States	Success, error, warning (low confidence, temporary lock), and information, always with text in addition to color

Note. Source: 12-ux-ui/design-system.md from the documentation repository.

Navigation map

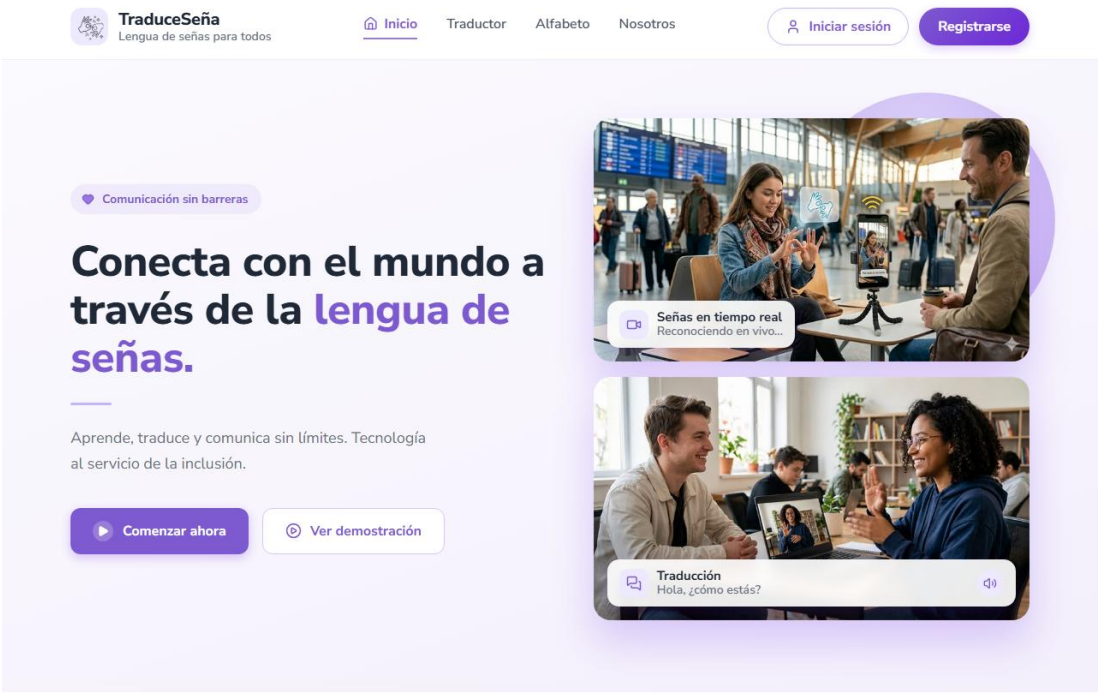
Table 2
System screens and their access

Screen	File in the frontend	Access
Home (landing), About us, Demos	homescreen/pages	Public
Sign in, Registration, Verify code	auth/pages	Public
Recover and change password	auth/pages	Public / authenticated
Translation	Translation/pages/TranslationScreen.tsx	Authenticated
Alphabet	Alphabet/pages/AlphabetScreen.tsx	Authenticated (public demo)
History	History/pages/HistoryScreen.tsx	Authenticated
Profile, Terms, Privacy	Profile/pages	Authenticated
Statistics	Stats/pages/StatsScreen.tsx	ADMIN
Administration panel, Training, Samples	Admin/pages	ADMIN

Prototypes of the main screens

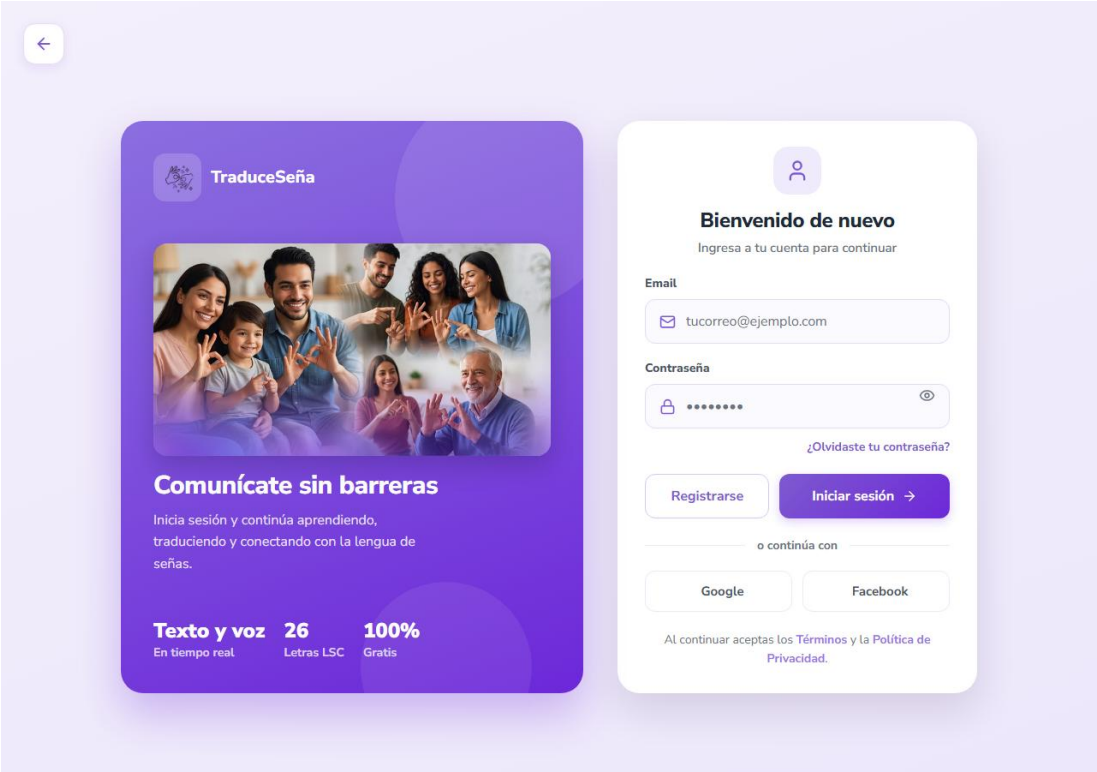
The following figures show the high-fidelity prototypes. They are the visual reference that the screens implemented in React Native follow.

Figure 1
Prototype — Public home page



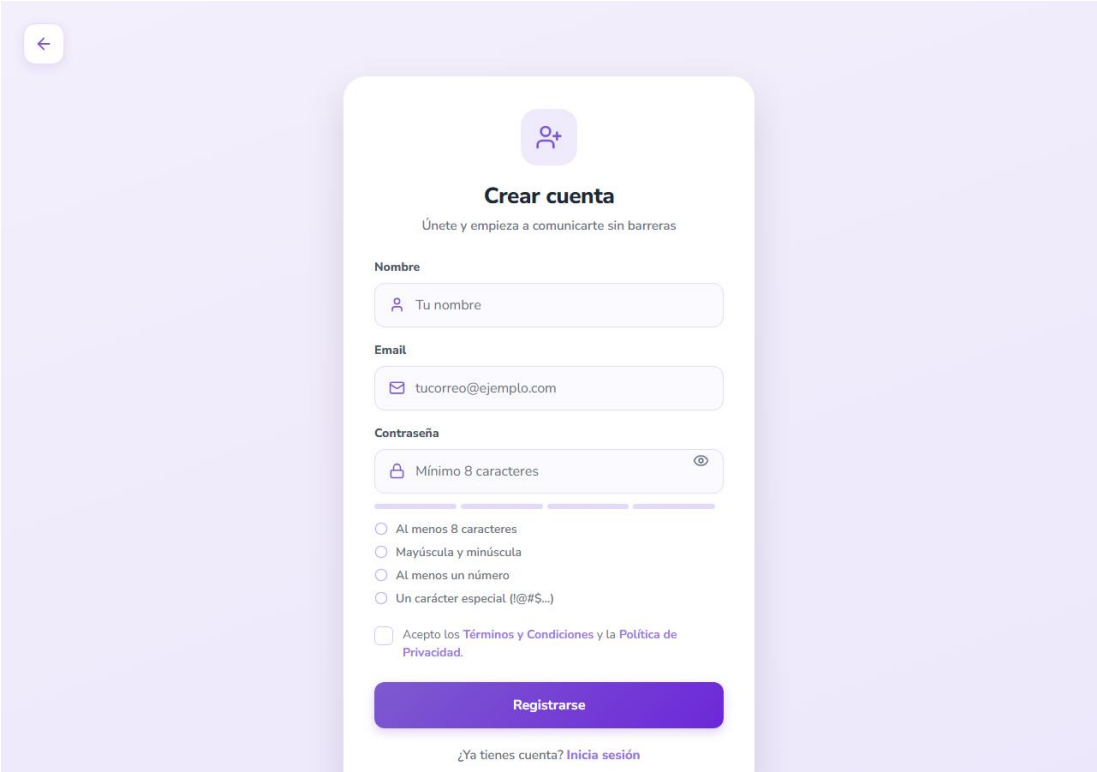
Note. High-fidelity project prototype (folder 12-ux-ui/mockup of the documentation repository).

Figure 2
Prototype — Sign-in



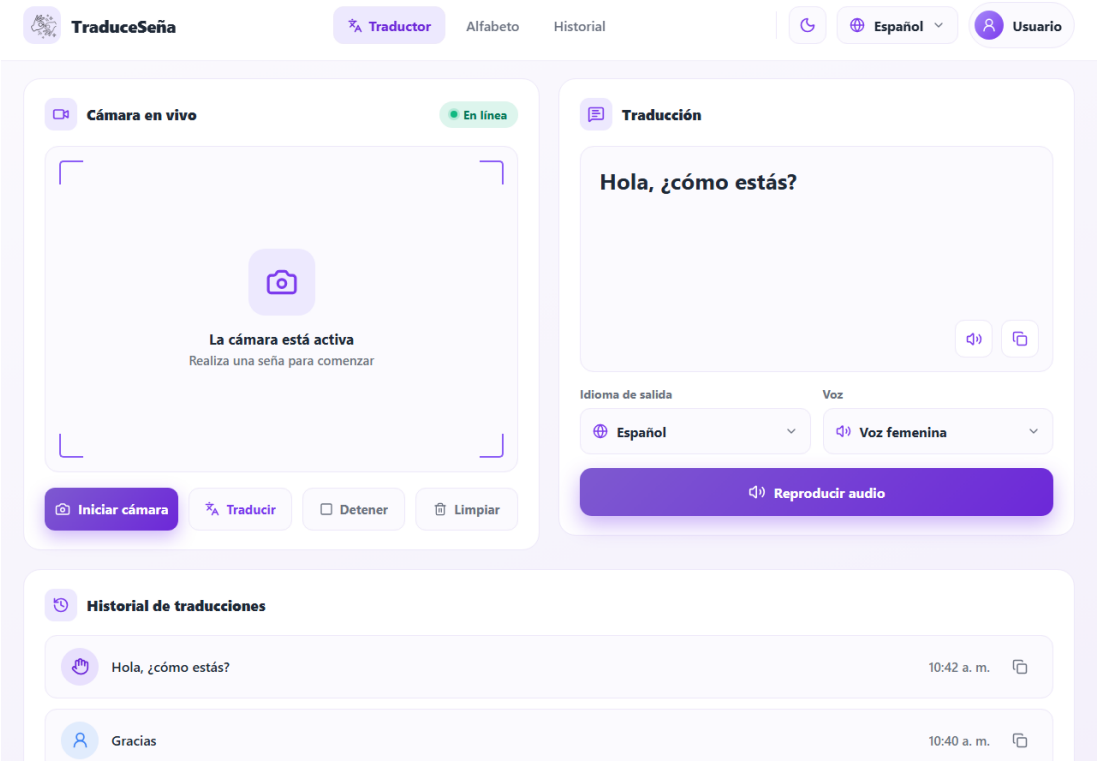
Note. High-fidelity project prototype (folder 12-ux-ui/mockup of the documentation repository).

Figure 3
Prototype — Account registration



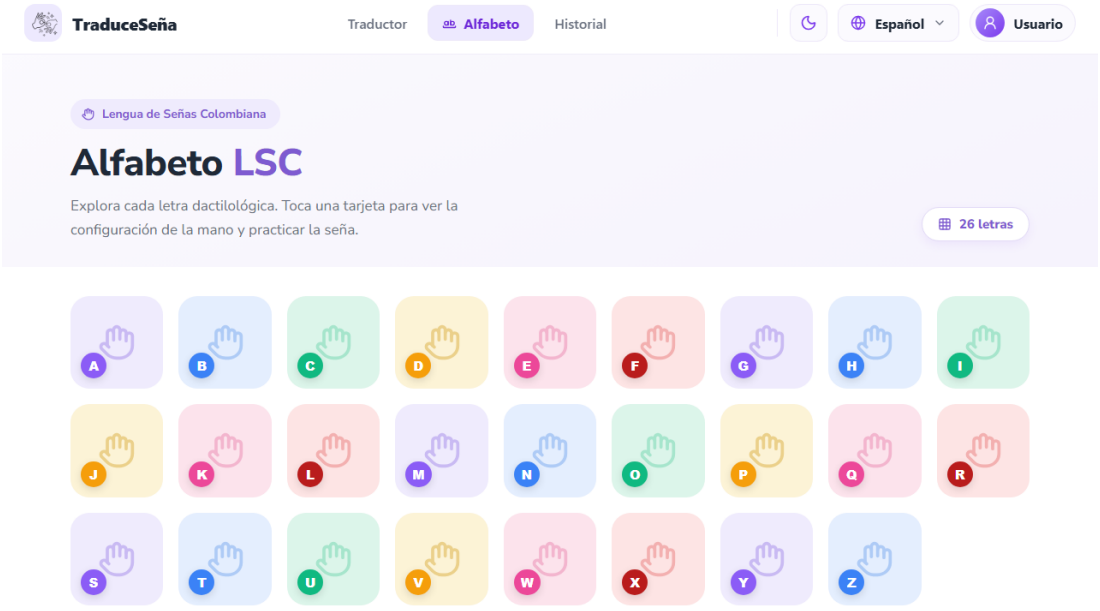
Note. High-fidelity project prototype (folder 12-ux-ui/mockup of the documentation repository).

Figure 4
Prototype — Live translation screen



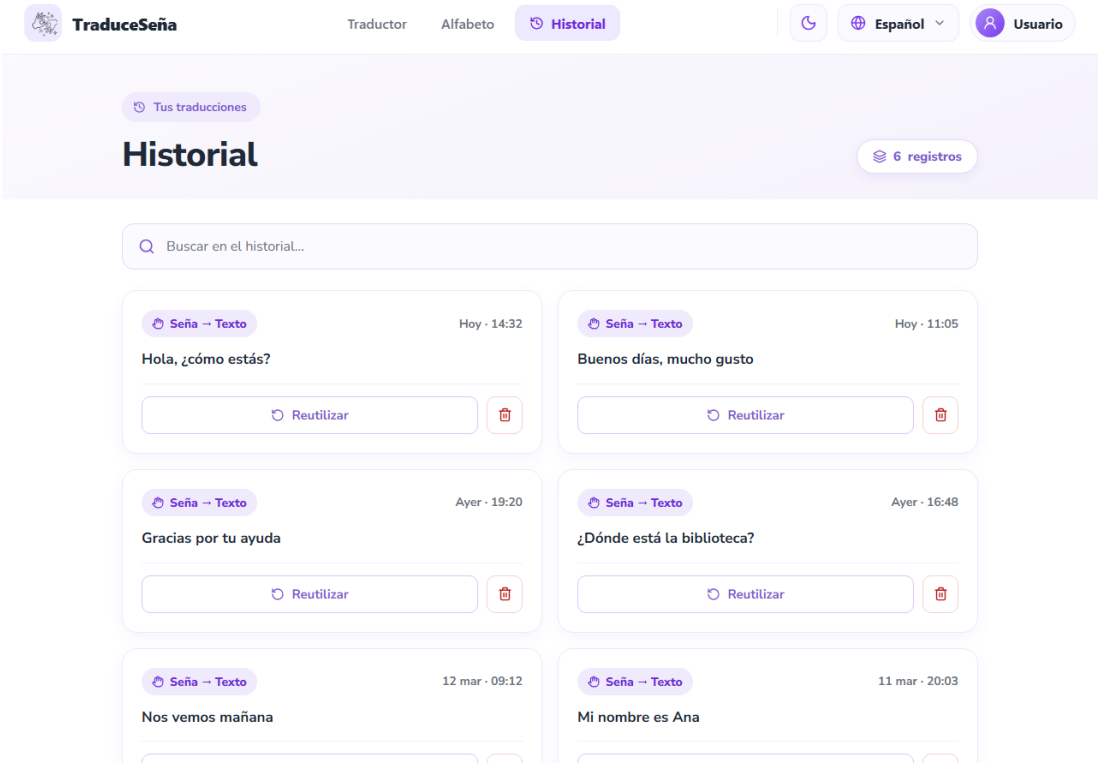
Note. High-fidelity project prototype (folder 12-ux-ui/mockup of the documentation repository).

Figure 5
Prototype — LSC alphabet



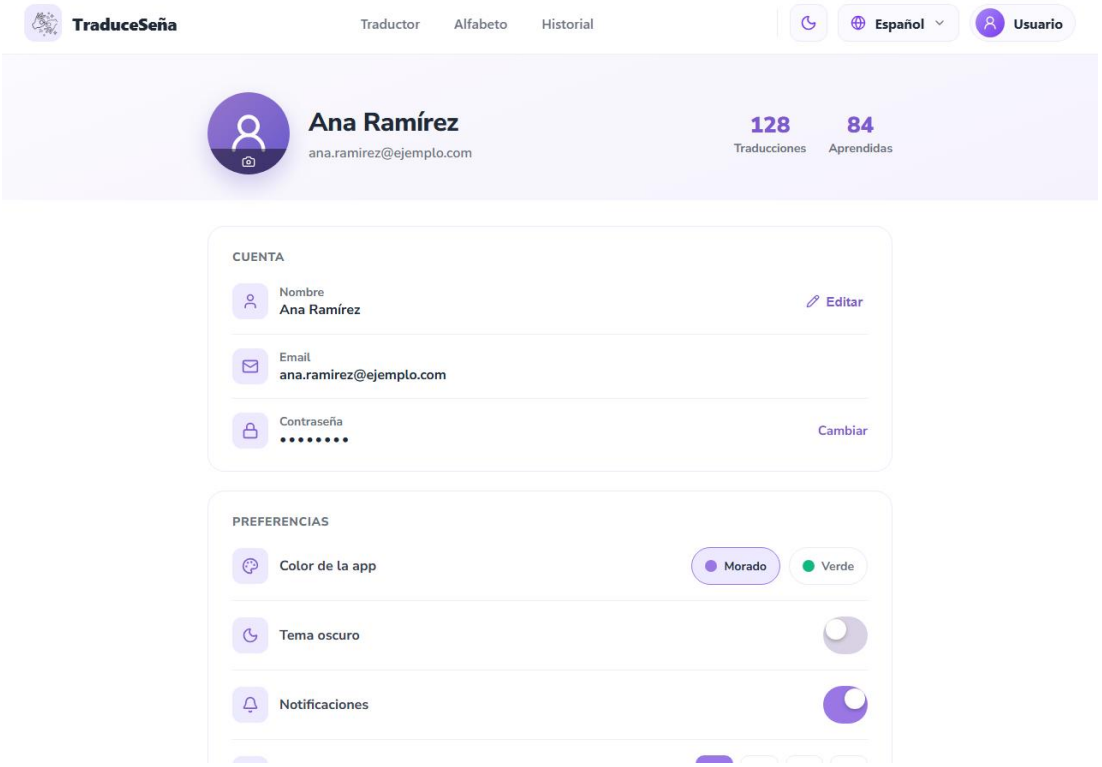
Note. High-fidelity project prototype (folder 12-ux-ui/mockup of the documentation repository).

Figure 6
Prototype — Translation history



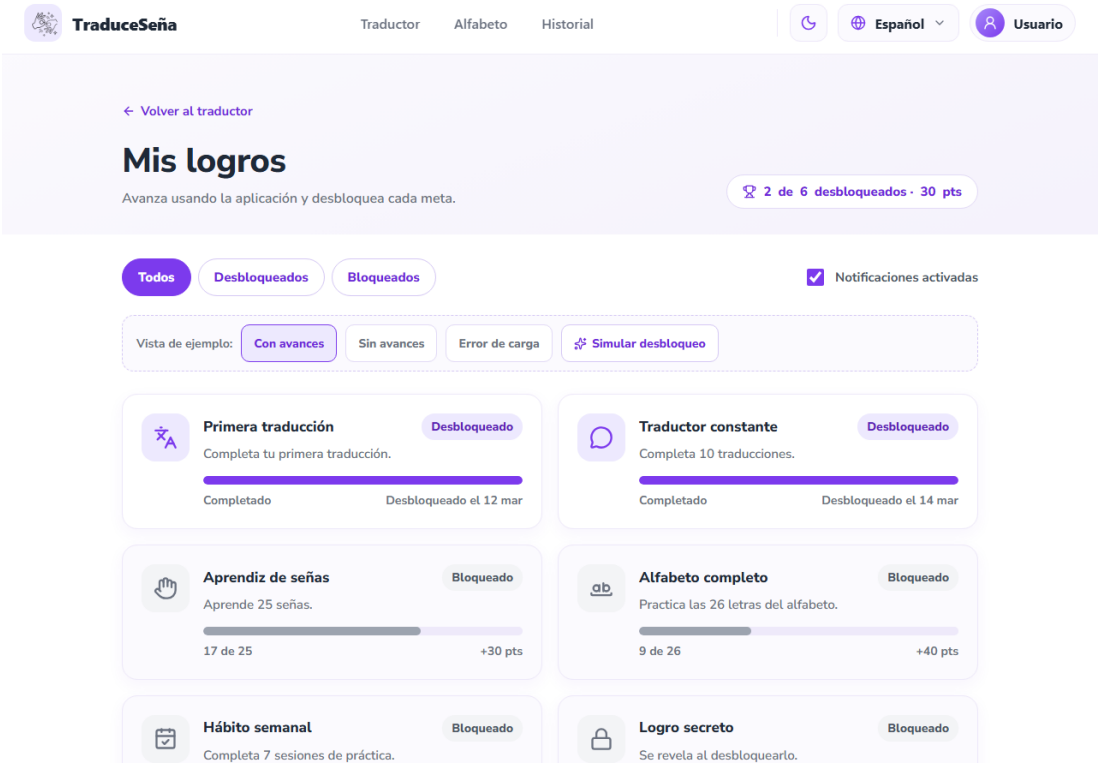
Note. High-fidelity project prototype (folder 12-ux-ui/mockup of the documentation repository).

Figure 7
Prototype — Profile management and preferences



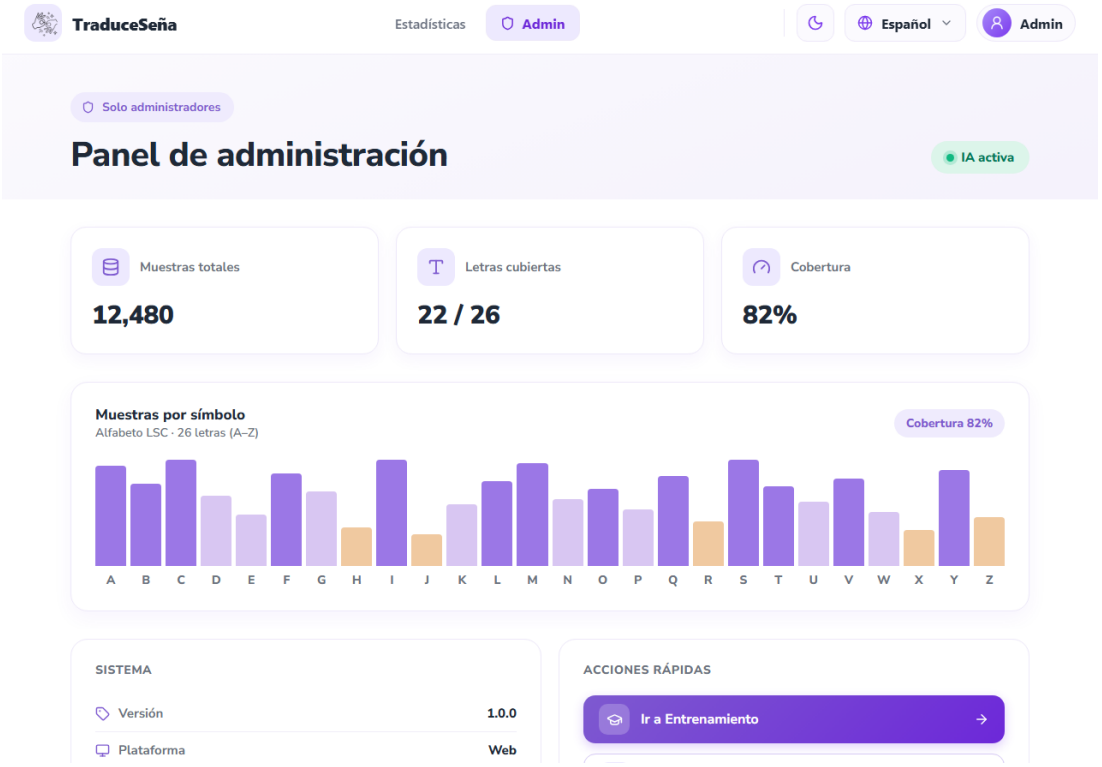
Note. High-fidelity project prototype (folder 12-ux-ui/mockup of the documentation repository).

Figure 8
Prototype — User achievements



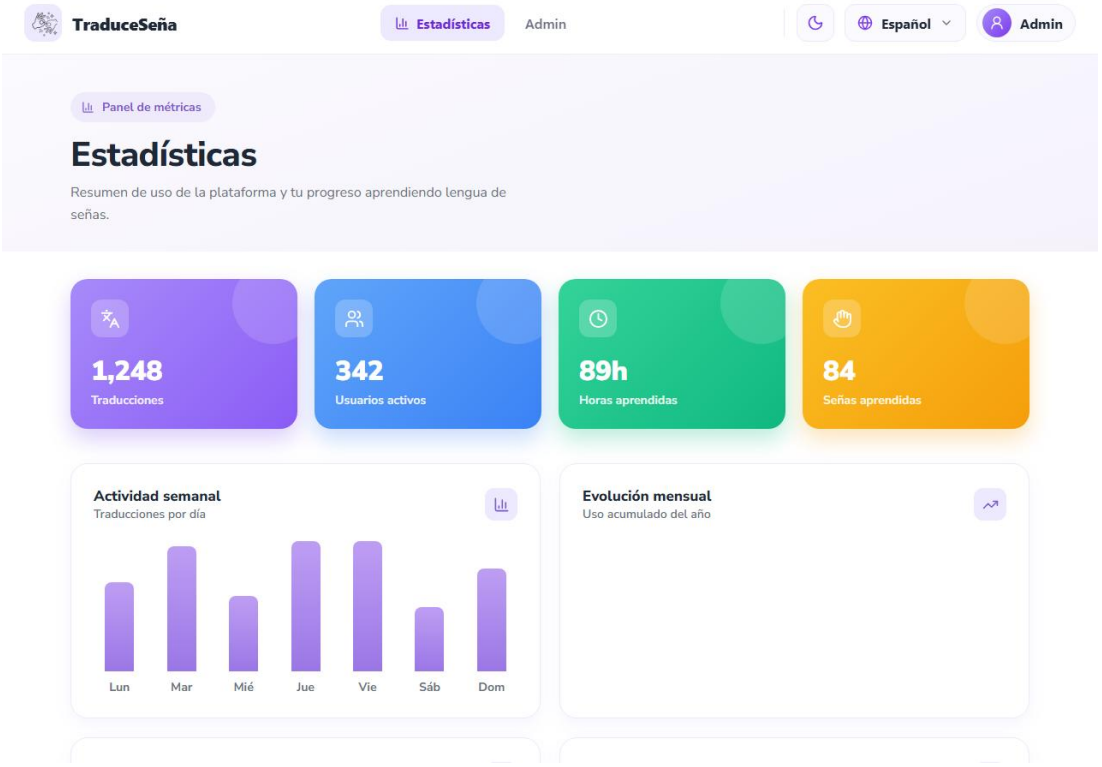
Note. High-fidelity project prototype (folder 12-ux-ui/mockup of the documentation repository).

Figure 9
Prototype — Administration panel



Note. High-fidelity project prototype (folder 12-ux-ui/mockup of the documentation repository).

Figure 10
Prototype — Usage statistics



Note. High-fidelity project prototype (folder 12-ux-ui/mockup of the documentation repository).

Figure 11
Prototype — Sample capture for training



Note. High-fidelity project prototype (folder 12-ux-ui/mockup of the documentation repository).

Architectural model

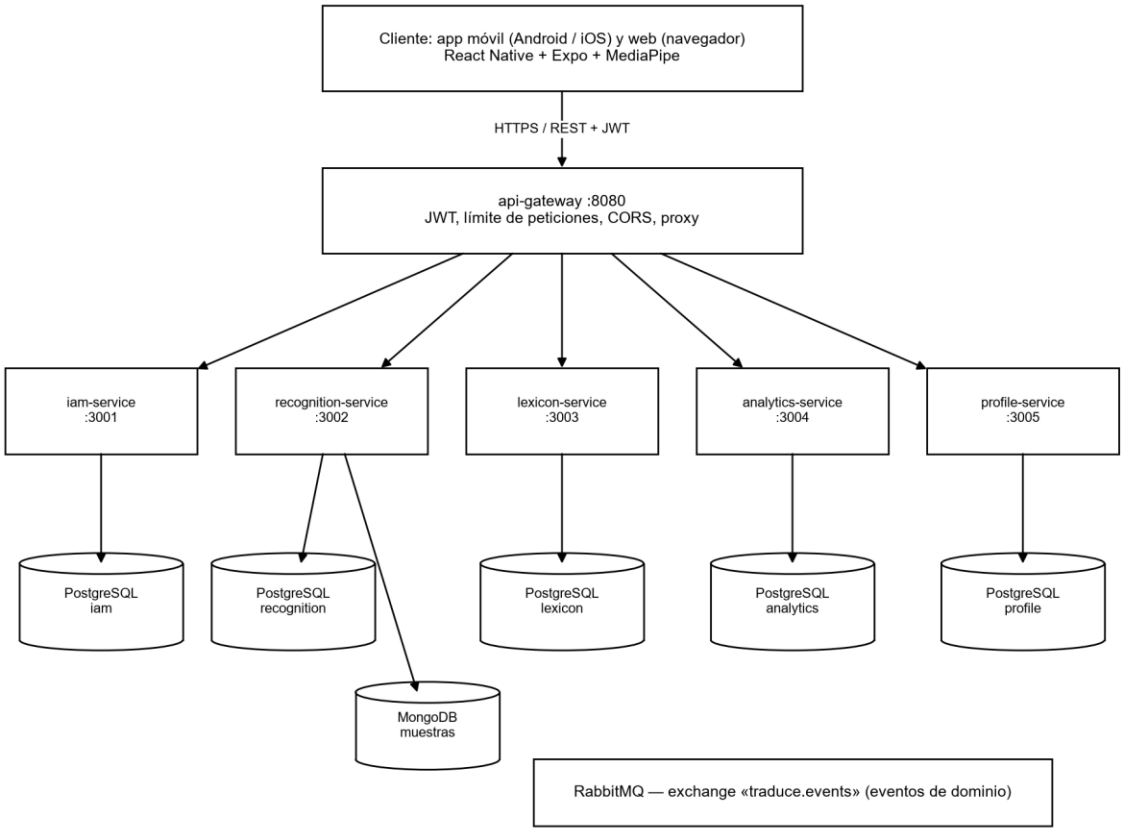
Architectural style

The backend follows a microservices architecture (Newman, 2021): five domain services behind an API Gateway. Each service has its own process, its own Dockerfile, and its own database; none reads another's database. Synchronous communication is REST through the gateway, and asynchronous communication consists of domain events over RabbitMQ.

Internally, each service applies the hexagonal or ports-and-adapters architecture (Cockburn, 2005; Martin, 2017): the domain and the use cases are at the center and do not know Express, PostgreSQL, or RabbitMQ.

Container view

Figure 12
Container view (C4 model, level 2)



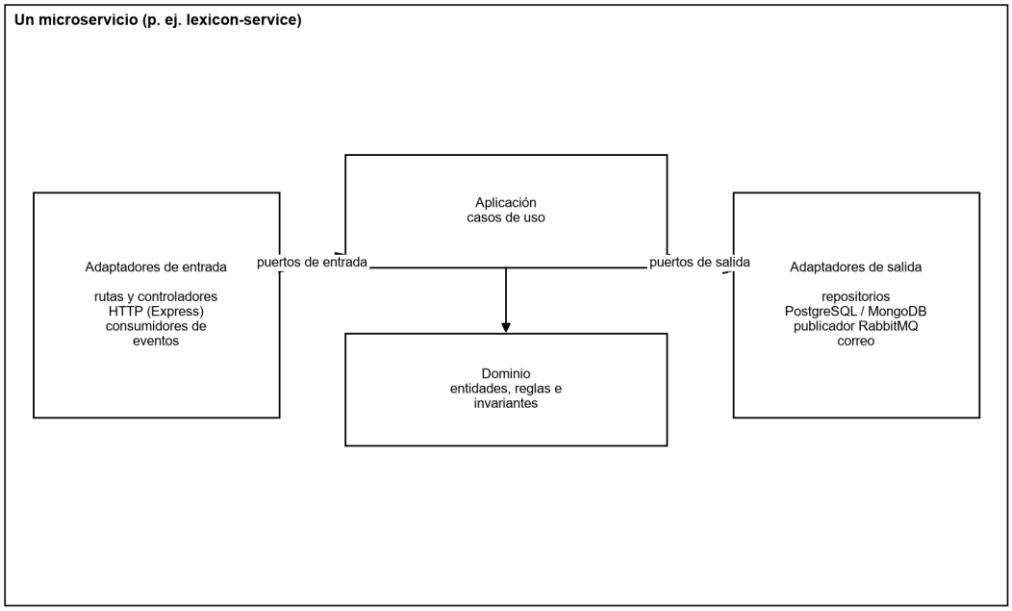
Note. Own elaboration based on the C4 model (Brown, 2018).

Table 3
Backend components

Component	Port	Responsibility	Storage
api-gateway	8080	Single entry point, JWT validation, request rate limit, CORS with allowlist, proxy	—
iam-service	3001	Registration, sign-in, email verification, roles and permissions, recovery	PostgreSQL iam
recognition-service	3002	Translations, sign templates, samples, and training	PostgreSQL recognition + MongoDB
lexicon-service	3003	Sign catalog, categories, locations, 3D alphabet, and media	PostgreSQL lexicon + .glb/.png files
analytics-service	3004	Usage events and aggregated statistics	PostgreSQL analytics
profile-service	3005	Profile, preferences, notifications, and achievements	PostgreSQL profile
@traduce/shared	—	Common package: configuration, JWT, HTTP middleware, and event bus	—

Internal architecture of each service

Figure 13
Hexagonal architecture of a microservice



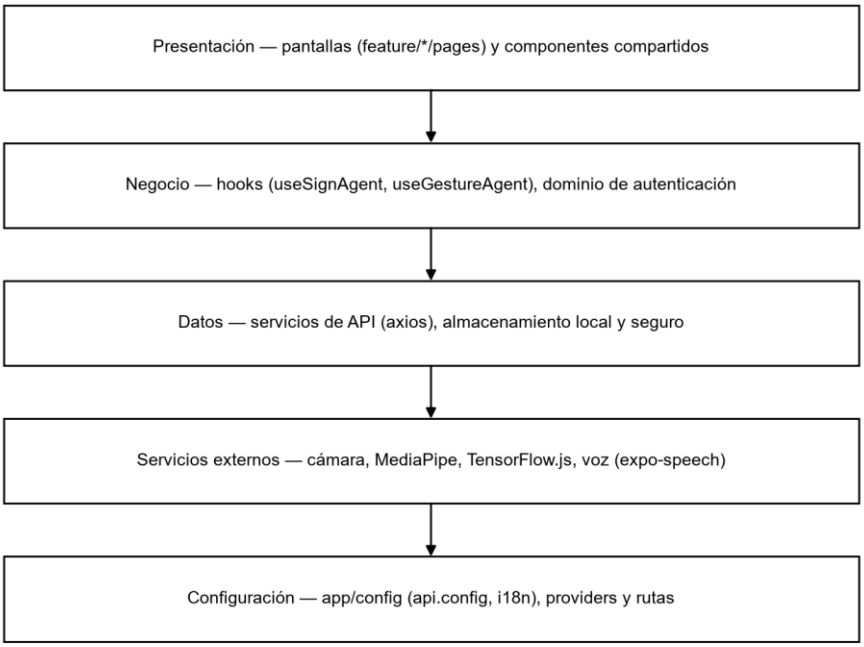
Note. Own elaboration.

```
services/lexicon-service/src/
├── domain/          entities, rules, and invariants
├── application/     use cases
├── ports/           input and output interfaces
├── adapters/
│   ├── inbound/http/   Express routes and controllers
│   └── outbound/       PostgreSQL repositories, RabbitMQ publisher
├── lexicon.module.ts   composition (dependency injection)
└── app.ts              service startup
```

Client architecture

The client is organized by feature (src/feature/Translation, Alphabet, auth, History, Profile, Stats, Admin, homescreen) and, within each one, by layer. Shared pieces live in src/shared and configuration in src/app.

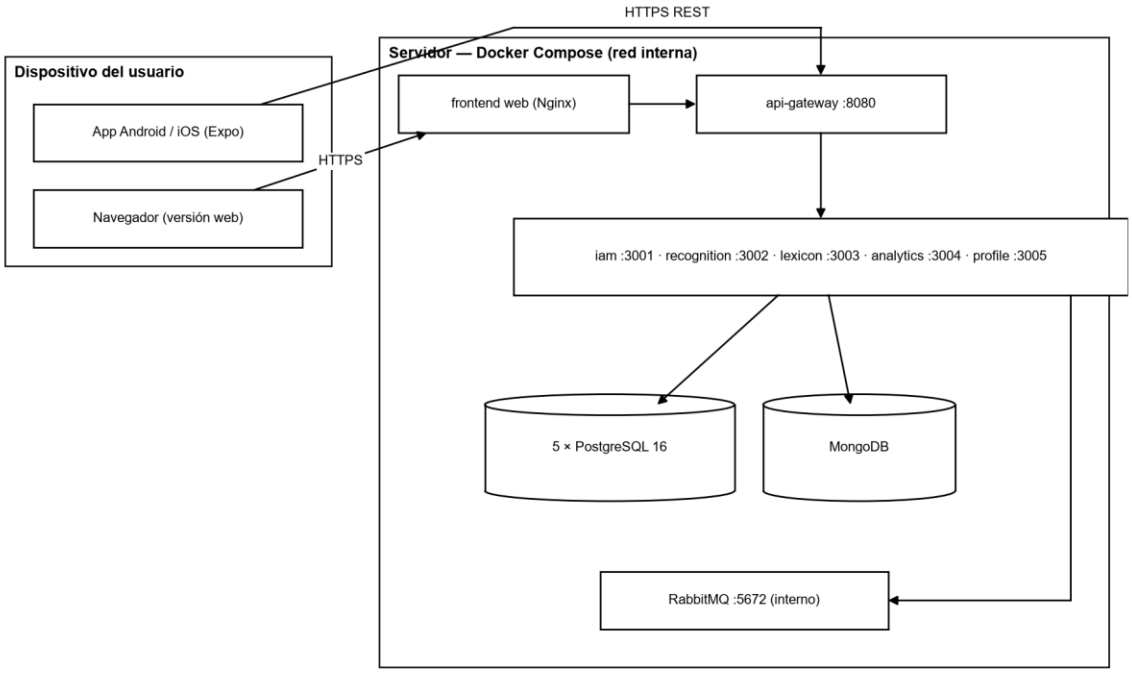
Figure 14
React Native client layers



Note. Own elaboration.

Deployment view

Figure 15
Deployment diagram



Note. Own elaboration.

Design patterns

The following table summarizes the patterns applied, where they are used, and what problem they solve. They are classified as architectural patterns, object-oriented design patterns (Gamma et al., 1994), and integration patterns.

Table 4
Design patterns applied

Pattern	Type	Where it is applied	Problem it solves
Ports and adapters (hexagonal)	Architectural	All microservices (domain/, ports/, adapters/)	Isolates business logic from technology
API Gateway	Architectural / integration	services/api-gateway	The client knows a single address; security is centralized
Database per service	Architectural	One PostgreSQL per service in docker-compose	Independence of deployment and schema
Publish-subscribe (domain events)	Integration	RabbitMQ bus in @traduce/shared; exchange traduce.events	Coordination without synchronous coupling

Pattern	Type	Where it is applied	Problem it solves
Repository	Design (Fowler, 2002)	PostgreSQL and MongoDB output adapters	Data access behind a domain interface
Dependency injection / composition root	Design	*.module.ts of each service	Use cases receive their ports already built; facilitates testing
Strategy	Behavior (GoF)	Client gesture providers: mediapipeProvider, tfjsProvider, mockProvider	Change the recognition engine without touching the screen
Adapter	Structural (GoF)	.web.ts files and native client files (gestureProvider.web.ts)	The same interface works on web and mobile
Chain of responsibility	Behavior (GoF)	Express middleware: CORS → rate limit → JWT → proxy	Each filter processes or rejects the request in order
Observer	Behavior (GoF)	React hooks (useSignAgent, useGestureAgent) and context providers	The interface reacts to changes in camera and session state
Data transfer object (DTO)	Design	Typed event contracts and HTTP responses	Stable contracts between layers and services
Retry with backoff	Resilience	subscribeWithRetry and event bus reconnection	Tolerates temporary RabbitMQ outages

SOLID principles

Table 5
Application of the SOLID principles

Principle	Application in the project
Single responsibility	Each use case is a class or function with a single task (for example, publishing a sign)
Open/closed	A new gesture provider is added by implementing the interface, without modifying the screen
Liskov substitution	The in-memory repositories in the tests substitute the PostgreSQL ones without changing behavior
Interface segregation	Small ports as needed (sign repository, event publisher, role reader)
Dependency inversion	The domain defines the ports; the infrastructure implements them and is injected into the module

Data model

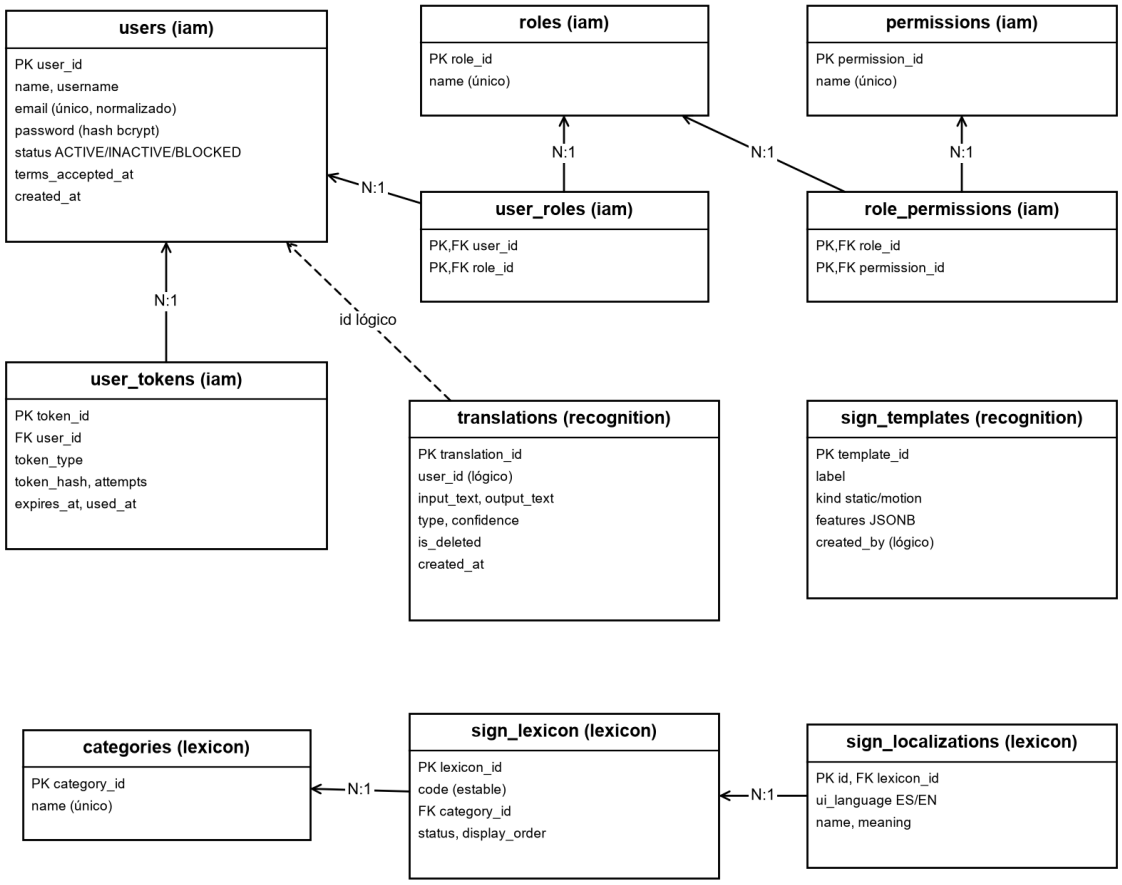
Model criteria

- **One database per service**; between services only logical identifiers are stored, without foreign keys.
- **Relational data in PostgreSQL** and gesture samples in **MongoDB**, because their structure varies and their volume grows separately.
- **Logical deletion** in the translation history (is_deleted); account deletion is physical, and the other services receive the iam.UserDeleted event.

- **Database constraints** for critical rules: normalized and unique email, valid states, codes with attempts ≥ 0, one location per language.
- Names in English and in snake_case; primary keys with _id suffix; dates as TIMESTAMPTZ.

Entity-relationship diagram

Figure 16
Entity-relationship model of the main services (implemented)



Note. Dashed lines indicate logical references between services, without foreign keys.

Summary data dictionary

Table 6
Data dictionary — users (iam-service)

Field	Type	Constraint	Description
user_id	SERIAL	PK	User identifier
name	VARCHAR(120)	NOT NULL	Full name

Field	Type	Constraint	Description
email	VARCHAR(255)	UNIQUE, normalized	Email in lowercase and without spaces
password	VARCHAR(255)	NOT NULL	bcrypt hash of the password
status	VARCHAR	ACTIVE / INACTIVE / BLOCKED	Account status
terms_accepted_at	TIMESTAMPTZ	—	Date terms were accepted (Law 1581)

Table 7
Data dictionary — user_tokens (iam-service)

Field	Type	Constraint	Description
token_id	SERIAL	PK	Identifier
user_id	INT	FK users	Code owner
token_type	VARCHAR(30)	EMAIL_VERIFICATION / PASSWORD_RESET / ACCOUNT_DELETION	Code use
token_hash	VARCHAR(255)	NOT NULL	Hash of the code; never in plain text
attempts	INT	≥ 0	Failed validation attempts
expires_at / used_at	TIMESTAMPTZ	—	Expiration and single use

Table 8
Data dictionary — translations (recognition-service)

Field	Type	Constraint	Description
translation_id	SERIAL	PK	Identifier
user_id	INT	Logical reference to iam	Owner of the translation
input_text / output_text	TEXT	NOT NULL	Recognized signs and resulting text
confidence	NUMERIC(4,3)	0 a 1	Recognition confidence
is_deleted	BOOLEAN	Default FALSE	Logical deletion

Table 9
Data dictionary — sign_templates (recognition-service)

Field	Type	Constraint	Description
template_id	SERIAL	PK	Identifier
label	VARCHAR(80)	NOT NULL	Sign represented (letter or word)
kind	VARCHAR(10)	static / motion / word	Static, motion, or word sign (ADR-011)
features	JSONB	NOT NULL	Normalized points of the template

Table 10
Data dictionary — sign_lexicon and sign_localizations (lexicon-service)

Field	Type	Constraint	Description
lexicon_id	SERIAL	PK	Sign identifier
code	VARCHAR(50)	Stable	Language-independent code
category_id	INT	FK categories	Sign category

Field	Type	Constraint	Description
status	VARCHAR(10)	Default ACTIVE	Only ACTIVE is visible to the user
ui_language	VARCHAR(2)	ES / EN, unique per sign	Language of the localization
name / meaning	VARCHAR	NOT NULL	Name and meaning in that language

Table 11
Data dictionary — Other services

Table	Service	Key constraint	Content
usage_event	analytics	unique event_id	Usage event by section, type, and signs
user_profiles	profile	logical PK user_id	User presentation data
user_preferences	profile	ES/EN, LIGHT/DARK	Language, theme, and notifications
notifications	profile	IN_APP / EMAIL channel (PUSH unused)	Visual-only notifications
achievements	profile	Meta > 0	Achievement catalog and progress; since migration 010 it includes challenge-point medals (ADR-010)
challenges	profile	Active definitions ordered by code	Daily challenge catalog (TRANSLATE_1, 3, 5 and 8, worth 5, 10, 15 and 25 points; ADR-009) and their texts in Spanish, English, French and Portuguese
user_challenges	profile	Unique per user and day	Snapshot of the challenge assigned to each user: target, progress and completion date
challenge_rewards	profile	One per assignment	Challenge points actually granted; their sum feeds the medals
admin_recipients, admin_report_runs	profile	One run per period	Recipients and control of the weekly and monthly email reports for administrators
gesture_samples	recognition (MongoDB)	Mandatory consent	Training samples

The code currently has 33 tables in PostgreSQL, spread across the five domain services, and one collection in MongoDB (gesture_samples). The complete reference model, with ten domains (identity, authorization, profile, lexicon, translation, AI, notifications, usage, gamification, and auditing), is maintained in DBML format in the documentation repository (06-data/traduce_senas.dbml). The tables shown here are those that already have a migration in the code. The current profile migrations run from 001 to 010 (the last three: 008 admin reports,

009 daily challenges and 010 challenge-point medals).

Two recent design rules do not change tables. History translations are deleted after 30 days (recognition-service), and the app shows the history grouped by sessions, one card per session. Each web screen also has its own URL. The weekly and monthly reports and the model reminder use the notification types ADMIN_WEEKLY_REPORT, ADMIN_MONTHLY_REPORT and ADMIN_MODEL_REMINDER, and are delivered as an in-app notice and by email (ADR-007). The daily challenge changes at midnight, Bogotá time, and only advances with saved translations.

Conclusions

The design combines an interface conceived first for accessibility with an architecture that separates responsibilities at two levels: between services (microservices with their own database) and within each service (hexagonal). The applied patterns make the code easy to test and change, and the data model protects critical rules with constraints in the database itself.

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